

Programming Assignment: Evaluating the GHS Distributed Minimum Spanning Tree Algorithm

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Theoretical Results

The GHS (Gallager-Humblet-Spira) algorithm is a distributed algorithm for constructing a minimum spanning tree (MST) in a weighted undirected graph. The algorithm operates asynchronously, allowing each node in the network to communicate with its neighbors to form fragments of the MST, which are progressively combined until the entire MST is constructed.

The communication cost and processing time of the GHS algorithm have been theoretically analyzed in the original paper [1]:

- **Communication Cost:** The algorithm requires $O(E + N \log N)$ messages, where E is the number of edges, and N is the number of nodes in the graph.
- **Processing Time:** The algorithm takes $O(N \log N)$ time in the best case, measured in terms of the number of message-passing rounds required to construct the MST.

The theoretical analysis suggests that the GHS algorithm is highly scalable, particularly in distributed systems with a large number of nodes and edges.

Algorithm Evaluation

The implementation of the GHS algorithm was conducted in Python, simulating the distributed environment using message-passing between processes to emulate nodes in a graph. Each node communicated with its neighbors asynchronously, and the fragments were combined based on the minimum-weight outgoing edges at each step.

The experiment focused on evaluating the algorithm’s performance using large-scale datasets, with particular attention to the communication cost (number of messages sent) and processing time (in terms of rounds) to reach the final MST.

The following libraries were used:

- NetworkX for graph creation and manipulation.
- Asyncio to simulate message passing between nodes.

Dataset

The dataset used for this evaluation was selected from the Stanford SNAP repository [2]. Specifically, the *p2p-Gnutella08* dataset was chosen because of its large size and realistic structure for simulating distributed systems. The graph contains:

- Number of Nodes: 6,301

- Number of Edges: 20,777

This dataset represents a large-scale network and provides a good test case for evaluating the scalability of the GHS algorithm in real-world conditions. The graph was preprocessed to ensure that it was connected and that all edge weights were positive.

Results

The experiments were conducted on graphs of varying sizes to evaluate the scalability of the GHS algorithm. Below is a summary of the communication cost (measured by the number of messages exchanged) and processing time (measured in seconds) for different numbers of partitions. Additionally, the average message count and processing time per segment are provided for comparison against theoretical complexity.

Partitions	Nodes per Parti- tion	Edges per Parti- tion	Total Msg.s	Total Proc. Time (Sec- onds)	Avg. Mes- sages	Avg Time (Sec- onds)
30	211	27	5,424	93.15	181	3.10
25	253	28	6,444	78.34	258	3.13
20	316	47	7,984	63.78	399	3.19
15	421	69	11,088	49.88	739	3.32

Table 1: Summary of experimental results for different partitions, including average metrics per subgraph.

The average message count and processing time per segment help assess the algorithm’s scalability across varying partitions. These averages can be compared against the theoretical time complexity of $O(N \log N)$ and message complexity of $O(E + N \log N)$.

Empirical and Theoretical Comparisons

Partitions	Nodes per Partition N	Edges per Partition E	Avg. Empirical Msg.s	Theoretical Msg.s $O(E + N \log N)$
30	211	27	181	27 + $211 \log 211 \approx 1,702$
25	253	28	258	28 + $253 \log 253 \approx 1,994$
20	316	47	399	47 + $316 \log 316 \approx 2,421$
15	421	69	739	69 + $421 \log 421 \approx 3,207$

Table 2: Average empirical message counts vs theoretical message complexity $O(E + N \log N)$

Partitions	Nodes per Partition N	Avg. Empirical Time (Seconds)	Theoretical Time $O(N \log N)$
30	211	3.10	$211 \log 211 \approx 1,675$
25	253	3.13	$253 \log 253 \approx 1,966$
20	316	3.19	$316 \log 316 \approx 2,374$
15	421	3.32	$421 \log 421 \approx 3,141$

Table 3: Average empirical processing time vs theoretical time complexity $O(N \log N)$

Conclusion

The comparison between empirical results and theoretical complexity leads to several observations:

- **Message Count:** The empirical average message counts are significantly lower than the theoretical upper bound $O(E + N \log N)$. The theoretical bound provides a limiting factor, while the empirical results suggest that the algorithm can perform more efficiently in practice. However, as the graph size increases (e.g., with fewer partitions and more nodes per partition), the gap between empirical and theoretical message counts narrows

slightly, indicating that the system scales closer to the upper bound in larger graphs. Additionally, potential synchronization issues in my GHS algorithm implementation, particularly with larger node counts, may have led to lower message counts and worsened scalability, implied by observed synchronization issues at higher node counts.

- **Processing Time:** The empirical processing times scale consistently with the theoretical complexity $O(N \log N)$. Although the empirical times (measured in seconds) are significantly lower than the values suggested by the theoretical bounds, the trend follows the expected growth rate. It is important to note that the theoretical time complexity is an abstract measure based on the number of communication rounds in the distributed system, rather than actual wall-clock time. Variations between empirical and theoretical processing times may be due to additional factors such as the overhead of simulating the distributed environment, asynchronous message-passing delays, and the performance of hardware components involved. These distributed system factors, such as network latency or message contention, could contribute to deviations from theoretical predictions.
- **Scalability:** The results confirm that the GHS algorithm scales well for large distributed systems, with message complexity and processing time growing in line with theoretical predictions. The observed gaps between empirical and theoretical values suggest that the algorithm benefits from inherent efficiencies during execution. However, as the network size increases, the increasing message counts and processing times indicate that the overhead associated with a distributed setup becomes more pronounced. The larger the partition, the more significant this overhead, especially in environments with high message synchronization or coordination costs.

In summary, the empirical results support the theoretical analysis, demonstrating that the GHS algorithm is scalable and efficient for constructing a minimum spanning tree in distributed systems. While the theoretical bounds provide a worst-case estimate, the algorithm performs better in practice, which is especially evident in the message counts for smaller partitions.

Bibliography

- [1] Gallager, R. G., Humblet, P. A., and Spira, P. M. (1983). A distributed algorithm for minimum-weight spanning trees. *ACM Transactions on Programming Languages and Systems*, 5(1), 66-77.
- [2] Leskovec, J., and Sosc, R. (2016). SNAP: A General Purpose Network Analysis and Graph Mining Library. <https://arxiv.org/abs/1606.07550>.